

EE0827 Final Project Specification

Embedded System Applications

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1 Overview

The final project is your opportunity to integrate the skills developed throughout the course into a complete embedded system. You will design, build, test, and document a system that demonstrates competency in hardware interfacing, real-time software architecture, and fault-tolerant design.

Team size: Individual or pairs (declare at proposal stage)

Duration: Weeks 11–14

2 Minimum Requirements

Every project must include **all** of the following:

Requirement	Course Reference	Evidence Required
Hardware interface with protection	Weeks 2–3	Schematic showing protection circuits
IMU sensor integration	Weeks 5, 9	Working demo + datasheet reference
Communication protocol (I2C, SPI, or robust UART)	Weeks 9, 10, or 11	Protocol waveform capture
Measurement evidence	All weeks	Oscilloscope screenshots, data tables
System documentation	—	Block diagram

Minimum Bar

Projects missing any minimum requirement cannot pass, regardless of other features. Verify all requirements before the final demo.

3 Milestones

Milestone	Deadline	Deliverables
M1: Proposal	End of W11	1-page proposal + block diagram + risk list
M2: Integration	End of W13	Working hardware demo

Milestone	Deadline	Deliverables
M3: Final Defense	W14	Live demo + code walkthrough + Q&A

3.1 M1: Proposal (Week 11)

Submit a 1-page proposal containing:

- **System description:** What does it do? (2-3 sentences)
- **Block diagram:** Hardware components and connections
- **Risk list:** Top 3 technical risks and mitigation strategies
- **Parts list:** Components needed (confirm availability)

3.2 M2: Integration Demo (Week 13)

Demonstrate to instructor:

- ☐ Hardware assembled and powered safely
- ☐ IMU sensor reading data correctly
- ☐ Basic system functionality working

3.3 M3: Final Defense (Week 14)

Live presentation (15-20 minutes) including:

- **Demo:** Working system with all minimum requirements
- **Code walkthrough:** Explain key sections of your implementation
- **Q&A:** Answer technical questions about design decisions

4 Recommended Project Tracks

Choose one of the following tracks, or propose your own (requires instructor approval).

4.1 Track A: Environmental Monitor

Description: Multi-sensor environmental monitoring station with data logging and alerts.

Suggested components:

- Temperature sensor (LM35 or thermistor)
- Humidity sensor (DHT22 or SHT30 via I2C)
- Light sensor (LDR or BH1750 via I2C)
- SD card module (SPI) for logging
- Buzzer or LED for alerts

Key challenges:

- Multiple I2C devices on shared bus
 - SPI interface for SD card (covered in Week 10)
 - Periodic logging with timestamps
 - Threshold-based alert system
-

4.2 Track B: Motor Control System

Description: Closed-loop motor control with speed/position feedback and safety features.

Suggested components:

- DC motor with encoder
- H-bridge driver (TB6612FNG or L298N)
- Current sense resistor
- Limit switches or home sensor

Key challenges:

- PID control loop in RTOS task
 - Encoder counting (timer input capture or interrupt)
 - Overcurrent and stall detection
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4.3 Track C: IMU & Sensor Fusion (Recommended for Advanced Students)

Description: Orientation tracking system using IMU with sensor fusion algorithm.

Suggested components:

Part	Interface	Notes
MPU6050	I2C	Budget option, 6-axis (accel + gyro)
ICM-20948	I2C/SPI	9-axis (accel + gyro + mag), recommended
BNO055	I2C	Has onboard fusion (easier but less learning)

Implementation levels:

Level	Algorithm	Complexity	Output
Basic	Complementary filter	Low	Pitch/roll angles
Intermediate	Mahony/Madgwick filter	Medium	Full 3D orientation
Advanced	Extended Kalman Filter	High	Orientation + bias estimation

Key challenges:

- High-speed I2C/SPI communication (use DMA for best results)
- Real-time filter computation
- Calibration (gyro bias, magnetometer hard/soft iron)
- Visualization (UART output to PC plotting tool)

Resources:

- Complementary filter: Simple weighted average of gyro integration and accelerometer angle
- Madgwick filter: [Open-source implementation](#)
- Calibration: Collect samples at rest, compute gyro bias as average

Stretch goals:

- Quaternion representation (avoids gimbal lock)
- Magnetometer fusion for absolute heading
- Motion detection / step counting
- Stream orientation to 3D visualization tool

Why This Track?

IMU sensor fusion combines multiple course topics: I2C/SPI protocols (W09), real-time filtering (W05), and DMA for high-throughput data (W10). It's directly applicable to robotics, drones, wearables, and VR controllers.

4.4 Track D: Custom Project

Propose your own project. Requirements:

- Must meet all minimum requirements
- Submit proposal by Week 11 for approval
- Discuss feasibility with instructor before committing

Past approved examples:

- Wireless sensor node (nRF24L01 + battery management)
- Audio spectrum analyzer (ADC + FFT + LED matrix)
- CAN bus interface for automotive diagnostics
- Servo-based pan-tilt camera mount with joystick control

5 Deliverables Summary

Item	Format	Submission
M1 Proposal	PDF (1 page + diagrams)	Upload to LMS
M2 Integration	Live demo	Instructor check in lab
Source Code	Git repository	Tag as final-project
Final Defense	In-person (W14)	Scheduled slot

6 Grading Rubric

Criterion	Points	Description
Minimum requirements met	40	All requirements demonstrated
Technical depth	25	Complexity appropriate to team size
Measurement evidence	15	Clear, relevant measurements with analysis
Documentation quality	10	Clear diagrams, readable code
Defense presentation	10	Clear explanation, answers questions

Total: 100 points

! Automatic Deductions

- Missing milestone: -10 points per milestone
- Late final submission: -5 points per day
- Missing minimum requirement: Cannot pass (max 50 points)

7 Resources

7.1 Hardware

- All components from weekly labs are available
- Additional sensors can be purchased (confirm with instructor)
- Breadboard prototyping is acceptable; PCB not required

7.2 Software

- STM32CubeIDE (as used in course)
- Starter code from weekly labs
- Open-source libraries allowed with citation

7.3 Documentation Templates

- Block diagram template: Draw.io or hand-drawn scan
- Report template: Follow lab report format (extended)

8 FAQ

Q: Can I use Arduino libraries? A: No. Use STM32 HAL/LL drivers. The goal is understanding, not shortcuts.

Q: What if my sensor doesn't arrive in time? A: Order early. If delayed, discuss alternatives with instructor immediately.

Q: Can I change tracks after proposal? A: Minor scope changes are fine. Major changes require re-approval.

Q: Do I need to implement Kalman filter? A: No. Complementary filter is sufficient for full marks. Kalman is a stretch goal.